

Roll No.

TPE-009

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
NEURAL NETWORKS AND MACHINE LEARNING

Time : Three Hours

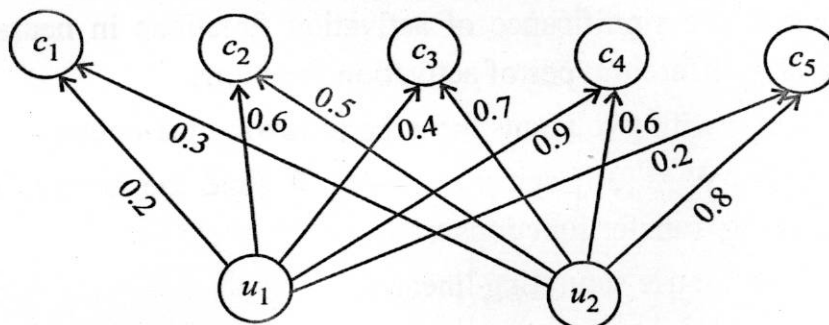
Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each sub-question carries 10 marks.

1. (a) Explain the structure of the neural networks in detail and compare it with the works of brain. (CO1)
(b) What is the significance of activation functions in neural networks ? Explain different types of activation functions. (CO1)
(c) A neuron with two inputs have the following parameters :
 $B = 0.8$, $W = [36]$ and $p = [4-6]$ T. Find the neuron output for the following transfer functions : (CO1)
 - (i) Symmetric saturating linear
 - (ii) Hard limit
 - (iii) Symmetrical hard limit
2. (a) Explain the differences between Associative and Autoassociative neural networks. (CO2)

P. T. O.

- (b) Write short notes on the following terms : (CO2)
- Supervised Learning
 - Reinforcement Learning
 - Unsupervised Learning
- (c) Discuss in detail about Hebb rule and delta learning rule. (CO2)
3. (a) Explain RBF and write the training steps of the RBF. (CO3)
- (b) With the help of proper mathematical equations, explain how SVM classifies a binary data. (CO3)
- (c) Differentiate between the classification approaches of RBF and SVM. (CO3)
4. (a) What are self-organizing maps ? Write the training steps of Kohonen SOM. (CO4)
- (b) What are Hamming nets ? Discuss in detail with proper diagrams. (CO4)
- (c) For a SOM is show in the following figure : (CO4)



Find the following :

- Using the square of the Euclidean distance, find the cluster unit C_j that is the closest to the input vector (0.2, 0.6).
- Using a learning rate of 0.2, find the new weights for unit C_j .

(3)

5. (a) Describe *two* examples, where neural networks can be applied in real world applications. (CO5, CO6)

(b) What is the significance of recurrent in RNNs ? Explain with detailed architecture. (CO5, CO6)

(c) Explain how Hopfield model is useful for artificial neural networks.

(CO5, CO6)

Roll No.

TEC-801

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022

COMPUTER ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw the basic structure of computer and explain with the help of a block diagram. (CO1)
- (b) Explain 16 bit instruction code for memory reference and register reference instructions. (CO1)
- (c) Design a 4 bit adder-subtractor circuit. (CO1)

P. T. O.

2. (a) Draw flowchart of Booth's algorithm for multiplication of two signed-2's complement numbers. (CO2, CO3)
- (b) Explain selective set and selective complement operations with the help of an example for each. (CO2, CO3)
- (c) Design BCD adder combinational circuit. (CO2, CO3)
3. (a) Classify the following instructions on the basis of memory reference, register reference and input-output reference instruction, for which hexadecimal codes are given. Also classify direct or indirect addressing modes in case of memory reference instruction : (CO3)
 - (i) 7800
 - (ii) F800
 - (iii) 3450
 - (iv) 9500
 - (v) F400
- (b) Differentiate between direct and indirect addressing modes. (CO3)
- (c) Draw and explain the flowchart for instruction cycle. (CO3)
4. (a) Write short notes on Cache Memory and Random Access Memory. (CO3, CO5, CO6)
- (b) Explain memory hierarchy with the help of a block diagram. (CO3, CO5, CO6)
- (c) Write a program to input one character. (CO3, CO5, CO6)

(3)

5. (a) What are *four* important steps in an instruction cycle. Explain instruction pipeline with the help of timing diagram for a program having 7 instructions with 4 steps each in an instruction. (CO2, CO4)
- (b) Draw flowchart for subtracting numbers in signed 2's complement representation. (CO2, CO4)
- (c) Draw multiport memory organization and crossbar switch organization interconnection structures. (CO2, CO4)